

Q.1: Define the term Ecology?

The term ecology was invented by the German biologist, Ernst Haeckel in 1866. Ecology is the Study of the interaction between the living components of the earth with the environment. Ecology for example is critical for understanding conservation, biodiversity issues, global climate change, human alterations of the environment, and the impact of pollutants on ecological systems.

Q.2: Write the major components of Ecology?

- 1) Species
- 2) Population
- 3) Community
- 4) Ecosystem
- 5) Biome
- 6) Biosphere

Q.3: Define the term Species?

A species is often defined as the largest group of organisms in which two individuals are Capable of reproducing fertile offspring, typically using sexual reproduction. For example, These happy face spiders look different, but since they can interbreed, they are considered the Same species: *Theridion grallator*.

Q.4: Define the term Community?

A biological community consists of all the populations of different species that live in a Given area. Community ecologists focus on interactions between populations and how these Interactions shape the community.

Q.5: Define the term Population?

A population is a group of organisms of the same species that live in the same area at the Same time. Population ecologists study the size, density, and structure of populations and how They change over time

Q.6: Define the term Ecosystem?

An ecosystem consists of all the organisms in an area, the community, and the abiotic factors That influences that community. Ecosystem ecologists often focus on flow of energy and Recycling of nutrients.

Q.7: Define the term Biome?

Biomes are very large ecological areas on the earth's surface, with fauna and flora (animals and plants) adapting to their environment. Biomes are often defined by abiotic factors such as Climate, relief, geology, soils and vegetation. A biome is NOT an ecosystem, although in a Way it can look like a massive ecosystem.

Q.8: Define the term Biosphere?

The biosphere is planet Earth, viewed as an ecological system. Ecologists working at the Biosphere level may study global patterns—for example, climate or species distribution—Interactions among ecosystems, and phenomena that affect the entire globe, such as climate Change

Q.9: Write sub-fields of Ecology?

Sub-fields of Ecology Are:

- 1) Organismal ecology
- 2) Population ecology

- 3) Community ecology
- 4) Ecosystem ecology
- 5) Landscape ecology
- 6) Global ecology

Q.10: Define Cell?

The cell is the basic structural, functional, and biological unit of all known living organisms. A Cell is the smallest unit of life that can replicate independently, and cells are often called the "building blocks of life". The study of cells is called cell biology. Without the cell, there is no life on the earth.

Q.11: Difference between Unicellular and Multicellular Organisms?

Unicellular Organisms:

In unicellular (single-celled) organisms, the single cell performs all life functions. It functions independently.

Multicellular Organisms:

Multicellular (many celled) organisms have various levels of organization within them. These organisms may perform all life functions together.

Q.12: Difference between Prokaryotes and Eukaryotes?

Prokaryotes:

Organisms that have no well-defined nucleus and organelles are called as Prokaryotes. A prokaryote is a single-celled organism that lacks a membrane bound nucleus, mitochondria, or any other membrane-bound organelle.

Eukaryotes:

Organisms that have well-defined nucleus and well-defined organelles are called as Eukaryotes. A eukaryote is any organism whose cells contain a nucleus and other organelles enclosed within membranes. Eukaryotes belong to the taxon Eukarya or Eukaryota.

Q.13: Define the basic concepts of Species?

1. Biological species concept
2. Ecological species concept
3. Morphological species concept
4. Phylogenetic species concept

Q.14: Define Population Characteristics?

- 1) Population size
- 2) Population growth
- 3) Population distribution
- 4) Population density

Q.15: Define the factors that will increase the size of the Population?

- 1) Reproductive fitness
- 2) Good weather
- 3) Medicine
- 4) Geographic isolation
- 5) No predators

Q.16: Define Dispersal Constraints?

Dispersal Constraints:

Dispersal constraints refers to both the movement of individuals (Animals, plants, fungi, bacteria, etc.) From their birth site to their breeding site ('natal

BT101 important subjective notes for mid term 2017

Dispersal'), as well as the movement from one breeding site to another ('breeding dispersal'). Dispersal is also used to describe the movement of propagules such as seeds and spores.

Q.17: Define the components of Ecosystem?

There are two major components of ecosystem are

1. Biotic components
2. Abiotic components

Q.18: Difference between Biotic and Abiotic Ecosystem?

Biotic Components:

Biotic components are the living things that shape an ecosystem.

Biotic components usually include:

- 1) Producers
- 2) Consumers
- 3) Decomposers

Abiotic Components:

Abiotic components are non-living chemical and physical parts of the environment that affect living organisms and the functioning of ecosystems. Abiotic components include physical conditions and non-living resources that affect living organisms in terms of growth, maintenance, and reproduction.

Q.19: Define the Physical and chemical components of Ecosystem?

Physical Components:

- 1) Water
- 2) Light
- 3) Temperature
- 4) Wind
- 5) Air
- 6) Soil

Chemical Components:

- 1) Oxygen
- 2) Carbon
- 3) Nitrogen
- 4) Phosphorus
- 5) Sulphur

Q.20: Define Ecological niche?

Ecological Niche:

An ecological niche is the role and position a species has in its environment; how it meets its needs for food and shelter, how it survives, and how it reproduces. A species' niche includes all of its interactions with the biotic and abiotic factors of its environment. Biotic factors are living things, while abiotic factors are nonliving things.

Q.21: Define Generalist species with example?

Those species that have broad range of niches are known as generalist species. A generalist species is able to thrive in a wide variety of environmental conditions and can make use of a variety of different resources.

Example:

Cockroach (survives in most climates, only needs water/moisture and a food source, only defense is responding to puffs of air), raccoon (wide diet, omnivore, lives in any area with trees, brush, or structures), are the examples of generalist species.

BT101 important subjective notes for mid term 2017

Q.22: Define Specialist species with example?

Those species that occupy narrow range of niches are known as specialist species.

A specialist species can thrive only in a narrow range of environmental conditions or has a Limited diet. Most organisms do not all fit neatly into either group, however.

Example:

Orchid mantis (colorful mantis with appendages like leaves, thrives only on orchids and in Tropics), sword-billed hummingbird (beak longer than body, co-evolved with flowers having Very long corollas and difficult getting food elsewhere), koala (lives almost entirely on Eucalyptus filling a niche that is toxic to most animals).

Q.23: Define Native Species?

Native species are those species that normally lives and thrives in a particular ecosystem. This Can include any species that developed with the surrounding habitat, and can be assisted by or Affected by a new species. Native species occurs naturally within a region. They either evolve in That region or arrive and become established in that region without human assistance.

Q.24: Define Endemic or Indigenous species with Example?

Endemic Species:

An endemic species is one whose habitat is restricted to a particular area. The term could refer to an animal, a plant, a fungus, or even a microorganism.

Example:

Blind Indus dolphin is only found in Indus river of Pakistan.

Indigenous Species:

A species is defined as indigenous to a given region or ecosystem if its presence in that region is the result of only natural process, with no human intervention.

Example:

Snow leopard is found in Pakistan, Nepal, India, and Russia and in some other countries of central Asia.

Q.25: Define Non-native species with example?

Non-native species are those species living outside its native distributional range, which has Arrived there by human activity, either deliberate or accidental. Non-native species can have Various effects on the local ecosystem.

Example:

Grass carp was introduced in Pakistan from china. It controls the growth of pond weeds as well as a great protein source.

Q.26: Impact of Non-native species on the Environment?

- 1) Non-native species can spread rapidly if they find a new more favorable niche.
- 2) In their new niches, these species often do not face the predators and diseases they faced before
- 3) They may be able to out-compete some native species in their new niches.

Q.27: Define Indicator species with Example?

An indicator species is an organism whose presence, absence or abundance reflects a specific Environmental condition. Indicator species can signal a change in the biological condition of a particular ecosystem, and thus may be used as a proxy to diagnose the health of an Ecosystem.

Example:

BT101 important subjective notes for mid term 2017

Birds are excellent biological indicators because they are found almost everywhere and are affected quickly by environmental changes such as loss or fragmentation of their habitats and introduction of chemical pesticides. The populations of many bird species are declining.

Q.28: Difference between Foundation vs. Keystone species?

Foundation Species:

A foundation species creates or provide a foundation of an ecosystem. For example, kelp is the foundation species for a kelp forest community. Without kelp, there would be no kelp forest.

Keystone Species:

Keystone species regulates and maintains that ecosystem. For example, sea otters are the keystone species of a kelp forest because they prey on urchins and if urchin populations get out of control then they will destroy all of the kelp. So otters maintain the habitat.

Q.29: Define Ecological Pyramid and its Classification?

An ecological pyramid is a graphical representation designed to show the biomass or bio Productivity at each trophic level in a given ecosystem. Biomass is the amount of living or Organic matter present in an organism.

Classification of ecological pyramid:

An ecological pyramid is classified into three main types:

1. Pyramid of number
2. Pyramid of biomass
3. Pyramid of energy

Q.30: Difference between Biomass and Productivity Pyramid?

Biomass Pyramid:

Biomass pyramids show how much biomass is present in the organisms at each trophic level.

Productivity Pyramid:

Productivity pyramids show the production in biomass.

Q.31: Define food Pyramid?

A food pyramid is a graphical representation of flow of energy through different trophic Levels of an ecosystem. The highest level an organism occupies on a food chain is termed as Trophic level. Energy supports the activities of all organisms in the ecosystem. It usually enters the ecosystem in the form of sunlight. It is incorporated into the chemical bonds of molecules within living and decaying tissues, and is eventually lost as heat.

Q.32: Define Habitat?

Actual place of an organism where it lives is called its habitat or environment. A habitat is an ecological or environmental area that is inhabited by a particular species of animal, plant, or Other type of organism. The term typically refers to the zone in which the organism lives and Where it can find food, shelter, protection and mates for reproduction. It is the natural environment in which an organism lives, or the physical environment that surrounds a Species population.

Q.33: Define Tolerance Range and range of optimum?

Tolerance Range:

Animals always live within a certain range of values, called the tolerance range, for any environmental factor. At either limit of the tolerance range, one or more essential functions Cease.

BT101 important subjective notes for mid term 2017

Range of optimum:

A certain range of values within the tolerance range is called the range of optimum.

Q.34: Define Limiting factor?

When an abiotic factor becomes out of an organism tolerance range then it is said to be a Limiting factor. For example

- 1) Oxygen
- 2) Water
- 3) Food

Q.35: Define Taxis?

Response of an organism towards any abiotic factor is known as taxis. Basically organisms Orient themselves to that response. For example, a response of organism to light is called Phototaxis.

Q.36: Define Food Chain?

The sequence of organisms through which energy moves in an ecosystem is a food chain. A Food chain starts with the primary energy source, usually the sun or boiling-hot deep sea Vents. The next link in the chain is an organism that makes its own food from the primary Energy source an example is photosynthetic plants that make their own food from sunlight (Using a process called photosynthesis) and chemosynthetic bacteria that make their food energy from chemicals in hydrothermal vents. These are called autotrophs or primary Producers.

Q.37: Define Primary Consumer?

Organisms that eat the autotrophs; these organisms are called herbivores or primary consumers an example is a rabbit that eats grass.

Q.38: Define Secondary Consumer?

The next link in the chain is animals that eat herbivores these are called secondary Consumers an example are a snake that eats rabbits.

Q.39: Define Tertiary Consumer?

These animals are eaten by larger predators an example is an owl that eats snakes.\

Q.40: Define Quaternary Consumer?

The tertiary consumers are eaten by quaternary consumers an example is a hawk that eats Owls

Q.41: Write Trophic levels of an Ecosystem?

The trophic level of an organism is the position it holds in a food chain.

- 1) Primary producers (organisms that make their own food from sunlight and/or Chemical energy from deep sea vents) are the base of every food chain - these Organisms are called autotrophs.
- 2) Primary consumers are animals that eat primary producers; they are also Called herbivores (plant-eaters).
- 3) Secondary consumers eat primary consumers. They are carnivores (meat-eaters) and omnivores (animals that eat both animals and plants).
- 4) Tertiary consumers eat secondary consumers.
- 5) Quaternary consumers eat tertiary consumers.
- 6) Food chains "end" with top predators, animals that have little or no natural enemies.

Q.42: Define Food Web?

A food web is the natural interconnection of food chains and generally a graphical

BT101 important subjective notes for mid term 2017

Representation (usually an image) of what-eats-what in an ecological community. Another name for food web is a consumer-resource system.

Q.43: Define the term Producers, Consumers, and Decomposers?

Producers:

Producers (autotrophs) are typically plants or algae. Plants and algae do not usually eat other organisms, but pull nutrients from the soil or the ocean and manufacture their own food Using photosynthesis. For this reason, they are called primary producers

Consumers:

Consumers (heterotrophs) are species that cannot manufacture their own food and need to consume other organisms. Animals that eat primary producers (like plants) are Called herbivores. Animals that eat other animals are called carnivores, and animals that eat both plant and other animals are called omnivores.

Decomposers:

Decomposers (detritivores) break down dead plant and animal material and wastes and release it again as energy and nutrients into the ecosystem for recycling. Decomposers, such as bacteria and fungi (mushrooms), feed on waste and dead matter, converting it into inorganic chemicals that can be recycled as mineral nutrients for plants to use again.

Q.44: Difference between Photoautotrophs and Chemoautotrophs?

Photoautotrophs:

Photoautotrophs are organisms that carry out photosynthesis. Using energy from sunlight, carbon dioxide and water are converted into organic materials to be used in cellular functions such as biosynthesis and respiration. In terrestrial environments plants are the predominant variety, while aquatic environments include a range of phototrophic organisms such as algae, Protists, and bacteria.

Chemoautotrophs:

Chemoautotrophs are organisms that derive their energy from the oxidation or breakdown of various inorganic or organic food substances in their environment. The process releases energy which supports metabolic activity in the organism.

Q.45: Define Phytoplankton and their types?

In oceans, lakes and slow-moving streams, phytoplanktons are the major producers. Phytoplanktons are simply microscopic floating plants. Fish and other aquatic animals eat the Phytoplankton as it floats through the water. Phytoplankton, a great source of food for a wide range of aquatic creatures including whales, shrimp, snails, and jellyfish. Phytoplanktons are of two types

1. Diatoms

2. Dinoflagellates

Q.46: Define the term Eutrophication?

Eutrophication is the enrichment of an ecosystem with chemical nutrients, typically Compounds containing nitrogen, phosphorus, or both. Eutrophication can be a natural process In lakes, occurring as they age through geological time. Human activities can accelerate the rate at which nutrients enter ecosystems (Fish, shellfish, birds and mammals).

Q.47: Define the term Red tide?

Red tide is a common name for a phenomenon known as an algal bloom (large concentrations Of aquatic microorganisms) when it is caused by a few species of dinoflagellates and the

BT101 important subjective notes for mid term 2017

Bloom takes on a red or brown color. Red tides are events in which estuarine, marine, or fresh Water algae accumulate rapidly in the water column, resulting in coloration of the surface Water. It is usually found in coastal areas. It kills many manatees every year.

Q.48: Define Photosynthesis?

Photosynthesis is a process used by plants and other organisms to convert light energy, Normally from the Sun, into chemical energy that can be later released to fuel the organisms' activities (energy transformation). This chemical energy is stored in carbohydrate molecules, Such as sugars, which are synthesized from carbon dioxide and water – hence the name Photosynthesis, from the Greek *phos*, "light", and *synthesis*, "putting together".

Q.49: Define the term Herbivores, Carnivores, and Omnivores?

Herbivores:

An herbivore is an animal that gets its energy from eating plants, and only plants. Omnivores can also eat parts of plants, but generally only the fruits and vegetables produced by fruit bearing plants. Many herbivores have special digestive systems that let them digest all kinds Of plants, including grasses.

Carnivores:

A carnivore is an animal that gets food from killing and eating other animals. Carnivores generally eat herbivores, but can eat omnivores, and occasionally other carnivores. Animals that eat other animals, like carnivores and omnivores are important to any ecosystem, Because they keep other species from getting overpopulated.

Omnivores:

An omnivore is a kind of animal that eats either other animals or plants. Some omnivores will hunt and eat their food, like carnivores, eating herbivores and other omnivores. Some others are scavengers and will eat dead matter. Many will eat eggs from other animals.

Q.50: Define the features of Primary Consumers?

1. Herbivores are animals that eat only plants.
2. They need a lot of energy to stay alive.
3. Many have to eat all day long to get enough energy.
4. Some herbivores have a special digestive system so they are able to eat/digest grasses.

Q.51: Define Detritivores?

An organism that feeds on and breaks down dead plant or animal matter, returning essential Nutrients to the ecosystem. Detritivores include microorganisms such as bacteria and Protists as well as larger organisms such as fungi, insects, worms, and isopod crustaceans. In a food Chain, detritivores are primary consumers.

Examples:

Typical detritivorous animals include millipedes, woodlice, dung flies, slugs, many Terrestrial worms, sea stars, sea cucumbers, fiddler crabs etc.

Q.52: Define Atmosphere?

The atmosphere is a mixture of nitrogen (78%), oxygen (21%), and other gases (1%) of the Air includes water vapor, carbon dioxide, and methane, all of which are called greenhouse Gases because they trap heat and thus warm the lower atmosphere. High above the planet, the atmosphere becomes thinner until it gradually reaches space. It is divided into five Layers. Most of the weather and clouds are found in the first layer.

Q.53: Define Hydrosphere?

BT101 important subjective notes for mid term 2017

Hydrosphere is a discontinuous layer of water at or near Earth's surface. It includes all liquid and frozen surface waters, groundwater held in soil and rock, and atmospheric water vapour.

Water is the most abundant substance at the surface of Earth. About 1.4 billion Cubic kilometres (326 million cubic miles) of water in liquid and frozen form make up the Oceans, lakes, streams, glaciers, and groundwater are found there. It is this enormous volume of water, in its various manifestations, that forms the discontinuous layer, enclosing much Of the terrestrial surface, known as the hydrosphere.

Q.54: Define Geosphere?

The geosphere is considered that portion of the Earth system that includes the Earth's interior, rocks and minerals, landforms and the processes that shape the Earth's surface. The Earth It (contrary to Christopher Columbus) is not a perfect sphere. It is what is called an oblate Spheroid, with a radius of 6,357 kilometers (km) from the Earth's center to the North Pole and 6,378 km from the center to the Equator.

Q.55: Define Biosphere?

The biosphere is the global sum of all ecosystems. It can also be termed as the zone of life on Earth, a closed system (apart from solar and cosmic radiation and heat from the Interior of the Earth), and largely self-regulating. By the most general bio Physiological definition, the biosphere is the global ecological system integrating all living Beings and their relationships, including their interaction with the elements of the lithosphere, geosphere, hydrosphere, and atmosphere. The biosphere is postulated to have evolved, beginning with a process of biopoesis (life created naturally from non-living matter, such as simple organic compounds) or biogenesis (life created from living matter), at least some 3.5 billion years ago.

Q.56: Difference between Biotic, low and high Potential?

Biotic potential

Ability of a population to grow under ideal conditions.

Low biotic potential

Large individuals, such as elephants and blue whales.

High biotic potential

Small individuals, such as bacteria and insects.

Q.57: Define Population Geography?

Population geography is a division of human geography. It is the study of the ways in which spatial variations in the distribution, composition, migration, and growth of populations are related to the nature of places.

Q.58: Define Carrying Capacity?

Carrying capacity as the maximum population of a given species that can survive indefinitely In a given environment. It was originally applied to relatively simple population environments Such as the number of sheep or cattle that could be maintained on grazing land Without degrading the land so that it could no longer support the animals.

Q.59: What is Secondary succession?

Secondary succession is one of the two types of ecological succession of plant life. As opposed to the first, primary succession, secondary succession is a process started by an event (e.g. forest fire, harvesting, and hurricane) that reduces an already established ecosystem (e.g. a forest or a

BT101 important subjective notes for mid term 2017

wheat field) to a smaller population of species, and as such secondary succession occurs on preexisting soil whereas primary succession usually occurs in a place lacking soil.

Q.60: Define primary succession?

The process by which a new community is propagated or initiated in an area where previously no life forms exist is termed as primary succession. Primary succession takes thousands of years to develop by the gradual accumulation of nutrients

Q.61: Define Climax community?

The relatively stable and uniform community developed by the pioneers through different succession stages is named as climax community. It is a final and mature stage of any community.

Q.62: Define Succession?

Succession is the process of change in the species structure of an ecological community over time. The time scale can be decades (for example, after a wildfire), or even millions of years after a mass extinction.

Q.63: Dif. B/w Autogenic succession and allogenic succession?

Allogenic succession is succession driven by the abiotic components of an ecosystem. An allogenic succession can be brought about in a number of ways which can include: Volcanic eruptions

- ♣ Meteor or comet strike
- ♣ Flooding
- ♣ Drought
- ♣ Earthquakes

Autogenic succession is driven by the biotic components of an ecosystem. Autogenic succession is brought about by internal biological processes.

- Shading by development of a tree canopy
- Accumulation of litter

Q.64: Define Species hotspots?

BT101 important subjective notes for mid term 2017

Species hotspots are geographic regions that support high level of species diversity. Tropical rainforests are one of important species hotspots region in the world because they are home to more than 50% species.

Q.65:What is Species dominance?

Species dominance refers to the biological supremacy of species that influence the ecosystem by controlling the energy flow in that ecosystem. Species dominance has an inverse relation with species diversity. Communities with low diversity usually have low evenness and are often high-dominance assemblages.

Q.66: Define Species diversity?

Species diversity refers to the measure of diversity in an ecological community. Species diversity takes into consideration species richness, which is the total number of different species in a community. It also takes into account evenness, which is the variation of abundance in individuals per species in a community.

Q.67: Define Species richness?

Species richness is the number of different species represented in an ecological community, landscape or region. Species richness is simply a count of species, and it does not take into account the abundances of the species or their relative abundance distributions. Species diversity takes into account both species richness and species evenness. Richness is a common measure of variety of life on earth.

Q.68: What Ecological indicators?

Species composition provides the essential description of the character of the vegetation at a site. Certain images are readily understood when major species are mentioned, eg pinon (Pinus sp.) - juniper (Juniperus sp.) woodland, and other common species are also presumed to be present as one becomes familiar with the vegetation.

Q.69: Define Species composition?

Species composition is all the different living organisms that develop or compose a community within an ecosystem. Species composition is an important tool for the determination of community structure of a specific place. Species composition refers to all of the living things within a specific environment.

Q.70: Define Community ecology?

BT101 important subjective notes for mid term 2017

Community ecology is the study of the interactions between species in communities on many spatial and temporal scales, including the distribution, structure, abundance, demography, and interactions between coexisting populations.

Q.71: Define Predation?

The most obvious form of species interaction is when one species eats another, predation being the technical term for this unfortunate fate. The predator is the organism that does the eating, the prey the one that gets eaten.

Q.72: Dif B/W hibernation winter sleep? 5 marks.

Hibernation

True hibernation occurs in small animals in which temperature to remain alive in cold environment. Animals remain inactive for weeks or even for months. temperature does not falls substantially and sleeping animals can wake and active quickly. It lasts for short period of time

Winter sleep

There are physiological aspects of sleep that are similar to hibernation, such as a reduced heart and breathing rate and lowered body temperature, but these changes are very slight compared to hibernation. Sleep is also pretty easy to break out of; if you are awakened from even deep sleep you can be fully awake within several minutes. Sleep is a mostly mental change and is primarily characterized by changes in brain activity.

Difference;

The key difference between hibernation and sleeping is that hibernating animals aren't just sleeping. They undergo drastic physiological changes. The most significant element of hibernation is a drop in body temperature, which can sometimes be as much as 18 C. The vital signs of a hibernating animal are very different from the vital signs of an awake animal. When an animal awakes from hibernation, it exhibits many signs of sleep deprivation and needs time over the next few days to recover.

Q.73: Define Hibernation?

Hibernation is a state of inactivity and metabolic depression in endotherms. Hibernation refers to a season of heterothermy that is characterized by low body temperature, slow breathing and heart rate, and low metabolic rate.

Q.74: What is Torpor?

BT101 important subjective notes for mid term 2017

Torpor is a state of decreased physiological activity in an animal, usually by a reduced body temperature and metabolic rate. Torpor is a mechanism that is under physiological control. Animals can regain its normal body activities in a short period of time after remaining in a state of torpor.

Q.75: Write down the Characteristics of Endotherms?

- ♣ High rate of metabolism
- ♣ Generate heat from body cellular activities
- ♣ Body insulation by fur or feather

Q.76: Write down the Characteristics of Ectotherms?

- ♣ Low metabolic rate
- ♣ Generate body heat from external environment
- ♣ Body insulation is very poor

Q.77: Define ectotherm?

An ectotherm is an organism in which internal physiological sources of heat are of relatively small or quite negligible importance in controlling body temperature. Such organisms (for example frogs) rely on environmental heat sources, which permit them to operate at very economical metabolic rates. Colloquially, some refer to these organisms as "cold blooded" though such a term is not technically correct, as the blood temperature of the organism varies with ambient environmental temperature.

Q.78: What is endotherm?

An endotherm is an organism that maintains its body at a metabolically favorable temperature, largely by the use of heat set free by its internal bodily functions instead of relying almost purely on ambient heat.

Q.79: Define Thermoregulation?

Thermoregulation is the process that allows the human body to maintain its core internal temperature. The state of having an even internal temperature is called homeostasis. All thermoregulation mechanisms are designed to return the body to homeostasis.

Q.80: What are Soil types?

BT101 important subjective notes for mid term 2017

Soil type usually refers to the different sizes of mineral particles in a particular sample. Soil is made up in part of finely ground rock particles, grouped according to size as sand and silt in addition to clay, organic material such as decomposed plant matter.

Q.81: What is weathering? 5 marks

Weathering is the breaking down of rocks, soil and minerals as well as wood and artificial materials through contact with the Earth's atmosphere, biota and waters. Weathering occurs in situ, roughly translated to: "with no movement", and thus should not be confused with erosion, which involves the movement of rocks and minerals by agents such as water, ice, snow, wind, waves and gravity and then being transported and deposited in other locations.

Weathering is classified into three types

1. Chemical weathering
2. Mechanical weathering
3. Biological weathering

1. Chemical weathering

Chemical weathering is the breakdown of rocks and minerals by changing their chemical composition by water, carbon dioxide, oxygen and other compounds. Chemical weathering

may change the size of pieces of rock materials, but definitely changes the composition. So one type of mineral changes into a different mineral. Chemical weathering works through chemical reactions that cause changes in the minerals.

2. Mechanical weathering

Mechanical weathering is the process of breaking big rocks into little ones. This process usually happens near the surface of the planet. Temperature also affects the land. The cool nights and hot days always cause things to expand and contract. That movement can cause rocks to crack and break apart.

3. Biological weathering

Biological weathering also means organic weathering. It is the disintegration of rocks as a result of the action by living organisms. Plant and animals have a significant effect on the rocks as they penetrate or burrow into the soil respectively. Biological weathering can work hand in hand with physical weathering by weakening rock or exposing it to the forces of physical or chemical weathering.

Q.82: Define planktons, phytoplankton and zooplankton?

Created by: Hafiza Mubeen, Muhammad Nouman, Usman Rubani and Shahzadi Shahzadi

BT101 important subjective notes for mid term 2017

Plankton;

A microscopic swimming and free floating organisms in water are termed as planktons.

Phytoplankton;

Microscopic free floating plants are called as phytoplankton. Greek for "drifting plants" which includes many types of Algae

Zooplankton

A microscopic drifting animal are known as zooplankton. Greek for "drifting animals".

Q.83: Discuss Marine water life zones? 5 marks

Marine life is found in three major life zones:

1. The coastal zone
2. Open
3. Ocean bottom

The coastal zone:

Coastal zone means the coastal waters (including the land therein and there under) and the adjacent shore lands (including the waters therein and there under), strongly influenced by each and in proximity to the shorelines of the several coastal states, and includes islands, transitional and intertidal areas, salt marshes, wetlands and beaches.

Open sea:

The part of ocean that is deep and away from the coastal zone is termed as Open Ocean. The main body of a sea or ocean, especially the part that is outside territorial waters and not enclosed, or partially enclosed, by land. On the basis of the light penetration open ocean is divided into three vertical zones.

1. Euphotic zone

The euphotic zone is the depth of the water in a lake or ocean that is exposed to such intensity of sunlight which designates compensation point, i.e. the intensity of light at which the rate of carbon dioxide uptake, or equivalently, the rate of oxygen production, is equal to the rate of carbon dioxide production, equivalently to the rate of oxygen consumption, reducing thus the net carbon dioxide assimilation to zero.

2. Bathyal zone

The Bathyal zone – (also known as midnight zone) is the part of the pelagic zone that extends from a depth of 1000 to 4000 meters (3300 to 13000 feet) below the ocean surface. It lies between the mesopelagic above and the abyss pelagic below. The average temperature hovers at about 4 °C (39 °F). Although larger by volume than the euphotic zone, the Bathyal zone is less densely populated. Sunlight does not reach this zone, meaning primary production, if any, is almost nonexistent. It is known as the midnight zone because of this feature.

3. Abyssal zone

The abyssal zone or abyss pelagic is a layer of the pelagic zone of the ocean. At depths of 4,000 to 6,000 meters (13,123 to 19,685 feet), this zone remains in perpetual darkness and never receives daylight. These regions are also characterized by continuous cold and lack of nutrients. The abyssal zone has temperatures around 2 °C to 3 °C (35 °F to 37 °F) through the large majority of its mass. It is the deeper part of the midnight zone which starts in the bathypelagic waters above.

Q.84: What is Species diversity?

Species diversity refers to the measure of diversity in an ecological community. Species diversity takes into consideration species richness, which is the total number of different species in a community. It also takes into account evenness, which is the variation of abundance in individuals per species in a community.

Q.85: What is the Importance of coastal wetlands?

These coastal aquatic systems provide important ecological and economic services.

- ⊞ Maintain water quality
- ⊞ Filter toxic pollutants, excess plant nutrients, and sediments,
- ⊞ They provide food
- , ⊞ Habitats and nursery sites for a variety of aquatic and terrestrial species.
- ⊞ They also reduce storm damage and coastal erosion by absorbing waves.
- ⊞ Supply timber and fuel wood to coastal communities.

Q.86: Define brackish water?

BT101 important subjective notes for mid term 2017

Brackish water is salt water and fresh water mixed together. It is saltier than fresh water, but not as salty as seawater. It may result from mixing of seawater with fresh water, as in estuaries, or it may occur in brackish fossil aquifers.

Q.87: Define Hypertrophic Lake?

Hypertrophic lakes are highly fertile and supersaturated in phosphorus and nitrogen. They have excessive phytoplankton growth which contributes to poor water clarity, poor suitability for recreational uses, and restricts the habitat for desirable fish. Lake Hakanoa and Lake Ngaroto are hypertrophic lakes.

Q.88: Define Mesotrophic Lake?

Mesotrophic lakes are lakes with an intermediate level of productivity. These lakes are commonly clear water lakes and ponds with beds of submerged aquatic plants and medium levels of nutrients.

Q.89: What is Eutrophic Lake?

A eutrophic body of water, commonly a lake or pond, has high biological productivity. Due to excessive nutrients, especially nitrogen and phosphorus, these water bodies are able to support an abundance of aquatic plants. Usually, the water body will be dominated either by aquatic plants or algae.

Q.90: Define Littoral zone?

The littoral zone adjoins the shore (and is thus the home of rooted plants) and extends down to a point called the light compensation level, or the depth at which the rate of photosynthesis equals the rate of respiration. Within the littoral zone producers are of two main types: rooted or benthic plants, and phytoplankton (plant plankton) or floating green plants, which are mostly algae.

Q.91: Define Limnetic Zone?

The limnetic zone includes all the waters beyond the littoral zone and down to the light compensation level. The limnetic zone derives its oxygen content from the photosynthetic activity of phytoplankton and from the atmosphere immediately over the lake's surface. Atmospheric source of oxygen becomes significant primarily when there is some surface disturbance of water caused by wind action or human activity.

Q.92: Define Profundal Zone?

The bottom and deep water area of a lake, which is beyond the depth of effective light penetration, is called the pro-fundal zone. In north-temperate latitudes, where winters are long

BT101 important subjective notes for mid term 2017

and severe, this zone has the warmest water (4°C) in the lake in winter and coldest water in summer.

Q.93: Define benthic zone?

The benthic zone is the ecological region at the lowest level of a body of water such as an ocean or a lake, including the sediment surface and some sub-surface layers. Organisms living in this zone are collectively called the benthos, e.g. the benthic invertebrate community, including crustaceans and polychaetes. The organisms generally live in close relationship with the substrate bottom and many are permanently attached to the bottom

Q.94: What is endotherm?

An endotherm is an organism that maintains its body at a metabolically favorable temperature, largely by the use of heat set free by its internal bodily functions instead of relying almost purely on ambient heat.